

Practice - Four-Ingredient Blending (Minimize Cost)

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:54

Blend 100 kg of feed from four ingredients ($X_1..X_4$ kg) at minimum cost (\$/kg 0.30, 0.25, 0.40, 0.20) meeting protein and fibre specs and an availability limit on ingredient 3. Four variables: no geometric plot, but every tableau chart still applies.

Model

Minimize $Z = 0.3X_1 + 0.25X_2 + 0.4X_3 + 0.2X_4$

subject to

[Protein (kg)] $0.2X_1 + 0.3X_2 + 0.4X_3 + 0.1X_4 \geq 22$

[Fibre (kg)] $0.05X_1 + 0.1X_2 + 0.02X_3 + 0.15X_4 \leq 8$

[Total (kg)] $X_1 + X_2 + X_3 + X_4 = 100$

[X3 availability] $X_3 \leq 30$

$X_1, X_2, X_3, X_4 \geq 0$

RESULT

OPTIMAL

$Z^* = 27$

$X_1 = 40$

$X_2 = 60$

$X_3 = 0$

$X_4 = 0$

iterations: 5 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 27$ (converted back: $Z = -Z'$)
- Alternative optima: non-basic X_4 has a zero coefficient in row 0 - bringing it in changes the solution but not Z .

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. The objective is minimisation → multiply by -1 and maximise $Z' = -Z$ (lecture slide 9). The final answer is converted back with $Z = -Z'$.
2. Protein (kg) (Type II, $\geq$): subtract surplus $X5 \geq 0$ (cost 0). Its column is -1 , so it cannot start in the basis → an artificial variable is needed.
3. Fibre (kg) (Type I, $\leq$): add slack $X6 \geq 0$ (unused resource, cost 0) → $X6$ is the starting basic variable of this row.
4. Total (kg) ($=$): already an equation, but it has no obvious basic variable → an artificial variable is needed.
5. X3 availability (Type I, $\leq$): add slack $X7 \geq 0$ (unused resource, cost 0) → $X7$ is the starting basic variable of this row.
6. Protein (kg): add artificial $X8 \geq 0$ with objective coefficient $-M$ (huge penalty) → it provides an immediately recognisable starting basic variable.
7. Total (kg): add artificial $X9 \geq 0$ with objective coefficient $-M$ (huge penalty) → it provides an immediately recognisable starting basic variable.

Maximize $Z' = -0.3X1 - 0.25X2 - 0.4X3 - 0.2X4 - MX8 - MX9$

subject to

$$0.2X1 + 0.3X2 + 0.4X3 + 0.1X4 - X5 + X8 = 22$$

$$0.05X1 + 0.1X2 + 0.02X3 + 0.15X4 + X6 = 8$$

$$X1 + X2 + X3 + X4 + X9 = 100$$

$$X3 + X7 = 30$$

$$X1, X2, X3, X4, X5, X6, X7, X8, X9 \geq 0$$

Standard form - coefficient matrix

	X1	X2	X3	X4	X5	X6	X7	X8	X9		RHS
max Z'	-0.3	-0.25	-0.4	-0.2	0	0	0	-M	-M		
Protein (kg)	0.2	0.3	0.4	0.1	-1	0	0	1	0	=	22
Fibre (kg)	0.05	0.1	0.02	0.15	0	1	0	0	0	=	8
Total (kg)	1	1	1	1	0	0	0	0	1	=	100
X3 availability	0	0	1	0	0	0	1	0	0	=	30

Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0.3	0.25	0.4	0.2	0	0	0	M	M	0
X8	1	0	0.2	0.3	0.4	0.1	-1	0	0	1	0	22
X6	2	0	0.05	0.1	0.02	0.15	0	1	0	0	0	8
X9	3	0	1	1	1	1	0	0	0	0	1	100
X7	4	0	0	0	1	0	0	0	1	0	0	30

Initial tableau. Basic variables: $X_8 = 22$, $X_6 = 8$, $X_9 = 100$, $X_7 = 30$. Non-basic (set to 0): X_1 , X_2 , X_3 , X_4 , X_5 . The values of the basic variables are read directly from the RHS column.

Initial tableau after Big-M preparation

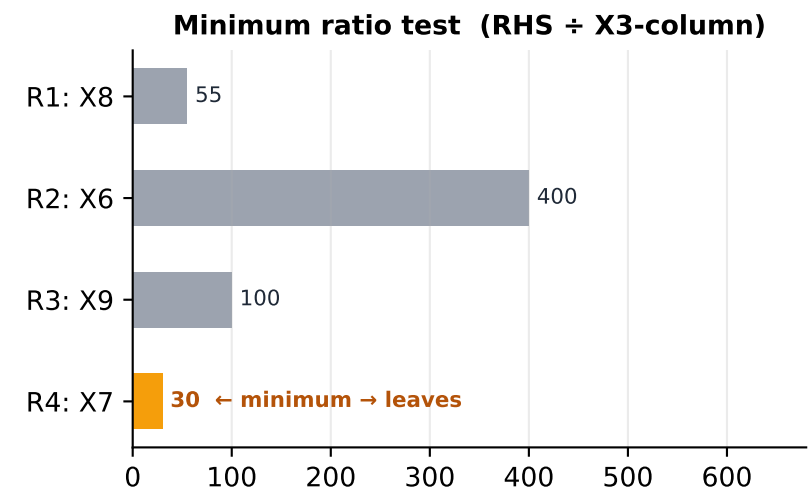
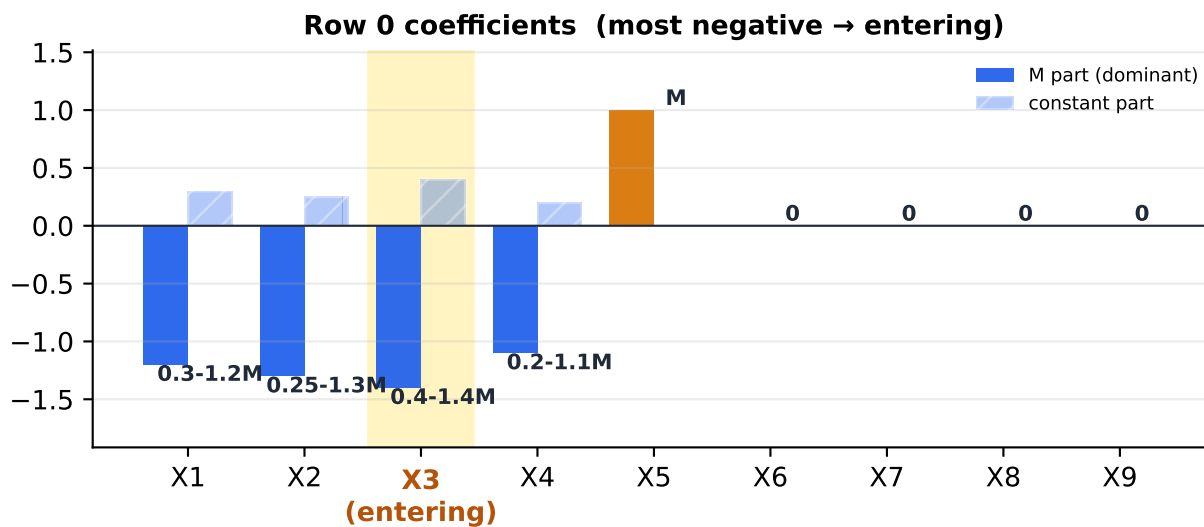
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0.3-1.2M	0.25-1.3M	0.4-1.4M	0.2-1.1M	M	0	0	0	0	-122M
X8	1	0	0.2	0.3	0.4	0.1	-1	0	0	1	0	22
X6	2	0	0.05	0.1	0.02	0.15	0	1	0	0	0	8
X9	3	0	1	1	1	1	0	0	0	0	1	100
X7	4	0	0	0	1	0	0	0	1	0	0	30

Big-M preparation. the artificial basic variable(s) had coefficient M in row 0. Subtracting M × their rows makes those coefficients zero, so row 0 is ready for the optimality test.

- Subtract M × R1 from R0 (make the coefficient of basic X8 zero)
- Subtract M × R3 from R0 (make the coefficient of basic X9 zero)

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.3-1.2M	0.25-1.3M	0.4-1.4M	0.2-1.1M	M	0	0	0	0	-122M	
X8	1	0	0.2	0.3	0.4	0.1	-1	0	0	1	0	22	55
X6	2	0	0.05	0.1	0.02	0.15	0	1	0	0	0	8	400
X9	3	0	1	1	1	1	0	0	0	0	1	100	100
X7	4	0	0	0	1	0	0	0	1	0	0	30	30



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X3 - it has the largest (in absolute value) negative coefficient in row 0 (0.4-1.4M).
- Minimum ratio test (RHS ÷ positive entries of the X3-column): R1: 55, R2: 400, R3: 100, R4: 30.
- Leaving basic variable: X7 (row 4, smallest ratio). Pivot number = 1.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Pivot number is already 1 - scaling not needed (normally divide the pivot row by the pivot number)

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.3-1.2M	0.25-1.3M	0.4-1.4M	0.2-1.1M	M	0	0	0	0	-122M	
X8	1	0	0.2	0.3	0.4	0.1	-1	0	0	1	0	22	55
X6	2	0	0.05	0.1	0.02	0.15	0	1	0	0	0	8	400
X9	3	0	1	1	1	1	0	0	0	0	1	100	100
X7	4	0	0	0	1	0	0	0	1	0	0	30	30

(b) X3 replaced X7 in the basis. Z = -12-80M

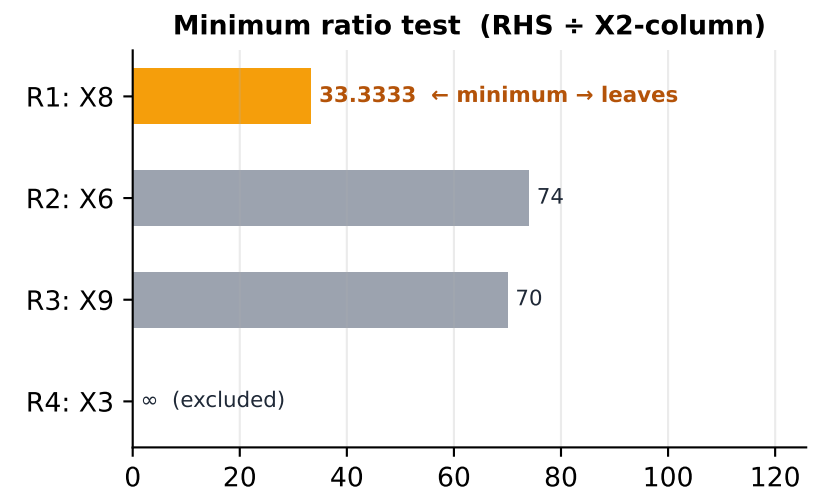
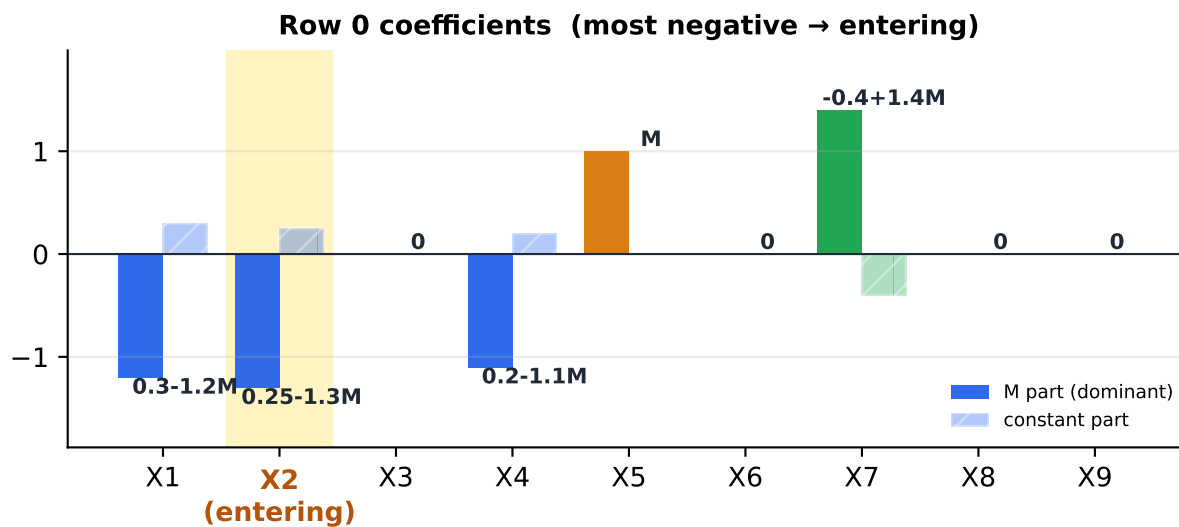
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0.3-1.2M	0.25-1.3M	0	0.2-1.1M	M	0	-0.4+1.4M	0	0	-12-80M
X8	1	0	0.2	0.3	0	0.1	-1	0	-0.4	1	0	10
X6	2	0	0.05	0.1	0	0.15	0	1	-0.02	0	0	7.4
X9	3	0	1	1	0	1	0	0	-1	0	1	70
X3	4	0	0	0	1	0	0	0	1	0	0	30

Row operations:

- Add $(-0.4+1.4M) \times R4$ to $R0$
- Subtract $0.4 \times R4$ from $R1$
- Subtract $0.02 \times R4$ from $R2$
- Subtract $1 \times R4$ from $R3$

Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.3-1.2M	0.25-1.3M	0	0.2-1.1M	M	0	-0.4+1.4M	0	0	-12-80M	
X8	1	0	0.2	0.3	0	0.1	-1	0	-0.4	1	0	10	33.3333
X6	2	0	0.05	0.1	0	0.15	0	1	-0.02	0	0	7.4	74
X9	3	0	1	1	0	1	0	0	-1	0	1	70	70
X3	4	0	0	0	1	0	0	0	1	0	0	30	∞



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 (0.25-1.3M).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: 33.3333, R2: 74, R3: 70, R4: ∞ .
- Leaving basic variable: X8 (row 1, smallest ratio). Pivot number = 0.3.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R1 by the pivot number 0.3

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.3-1.2M	0.25-1.3M	0	0.2-1.1M	M	0	-0.4+1.4M	0	0	-12-80M	
X8	1	0	0.6667	1	0	0.3333	-3.3333	0	-1.3333	3.3333	0	33.3333	33.3333
X6	2	0	0.05	0.1	0	0.15	0	1	-0.02	0	0	7.4	74
X9	3	0	1	1	0	1	0	0	-1	0	1	70	70
X3	4	0	0	0	1	0	0	0	1	0	0	30	∞

(b) X2 replaced X8 in the basis. Z = -20.3333-36.6667M

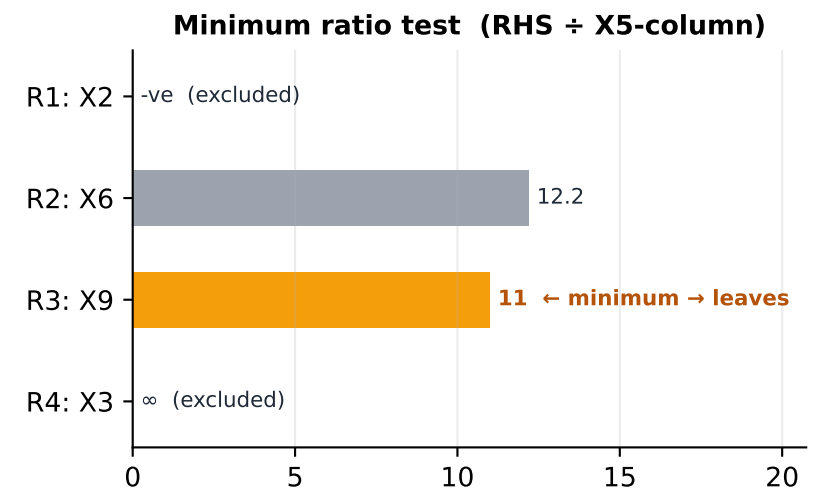
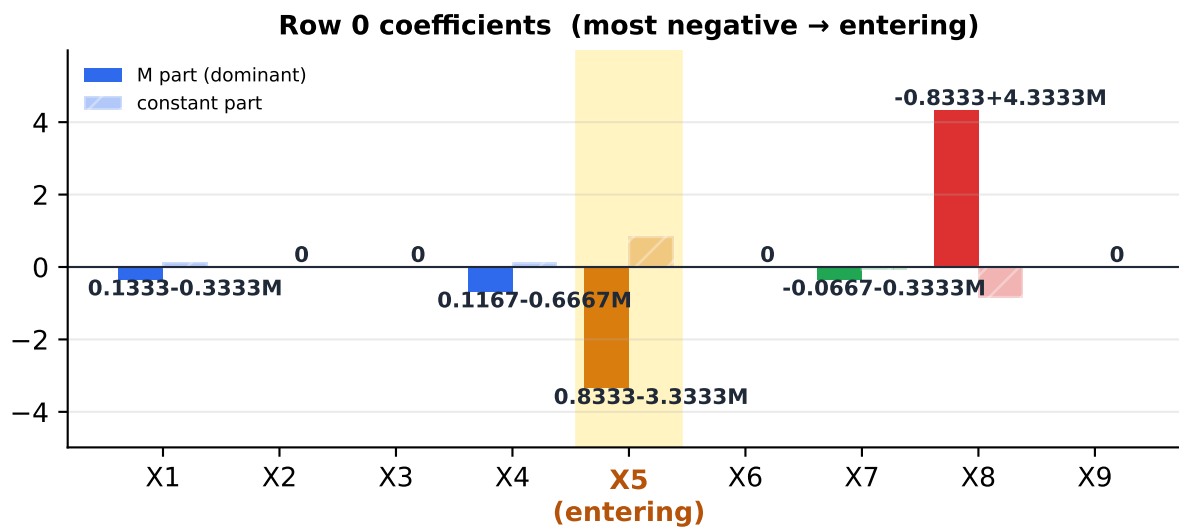
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0.1333-0.3333M	0	0	0.1167-0.6667M	0.3333-3.3333M	0	-0.0667-0.3333M	0.3333+4.3333M	0	-20.3333-36.6667M
X2	1	0	0.6667	1	0	0.3333	-3.3333	0	-1.3333	3.3333	0	33.3333
X6	2	0	-0.0167	0	0	0.1167	0.3333	1	0.1133	-0.3333	0	4.0667
X9	3	0	0.3333	0	0	0.6667	3.3333	0	0.3333	-3.3333	1	36.6667
X3	4	0	0	0	1	0	0	0	1	0	0	30

Row operations:

- Add $(-0.25+1.3M) \times R1$ to R0
- Subtract $0.1 \times R1$ from R2
- Subtract $1 \times R1$ from R3

Iteration 3: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	$0.1333-0.3333M$	0	0	$0.1167-0.6667M$	$0.8333-3.3333M$	0	$-0.0667-0.3333M$	$0.8333+4.3333M$	0	$-20.3333-36.6667M$	
X2	1	0	0.6667	1	0	0.3333	-3.3333	0	-1.3333	3.3333	0	33.3333	-ve
X6	2	0	-0.0167	0	0	0.1167	0.3333	1	0.1133	-0.3333	0	4.0667	12.2
X9	3	0	0.3333	0	0	0.6667	3.3333	0	0.3333	-3.3333	1	36.6667	11
X3	4	0	0	0	1	0	0	0	1	0	0	30	∞



Iteration 3 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X5 - it has the largest (in absolute value) negative coefficient in row 0 ($0.8333-3.3333M$).
- Minimum ratio test (RHS ÷ positive entries of the X5-column): R1: -ve, R2: 12.2, R3: 11, R4: ∞ .
- Leaving basic variable: X9 (row 3, smallest ratio). Pivot number = 3.3333.

Iteration 3: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R3 by the pivot number 3.3333

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.1333-0.3333	0	0	0.1167-0.6667	0.8333-3.3333	0	-0.0667-0.3333	0.8333+4.3333	0	-20.3333-36.6667	
X2	1	0	0.6667	1	0	0.3333	-3.3333	0	-1.3333	3.3333	0	33.3333	-ve
X6	2	0	-0.0167	0	0	0.1167	0.3333	1	0.1133	-0.3333	0	4.0667	12.2
X9	3	0	0.1	0	0	0.2	1	0	0.1	-1	0.3	11	11
X3	4	0	0	0	1	0	0	0	1	0	0	30	∞

(b) X5 replaced X9 in the basis. Z = -29.5

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0.05	0	0	-0.05	0	0	-0.15	M	-0.25+M	-29.5
X2	1	0	1	1	0	1	0	0	-1	0	1	70
X6	2	0	-0.05	0	0	0.05	0	1	0.08	0	-0.1	0.4
X5	3	0	0.1	0	0	0.2	1	0	0.1	-1	0.3	11
X3	4	0	0	0	1	0	0	0	1	0	0	30

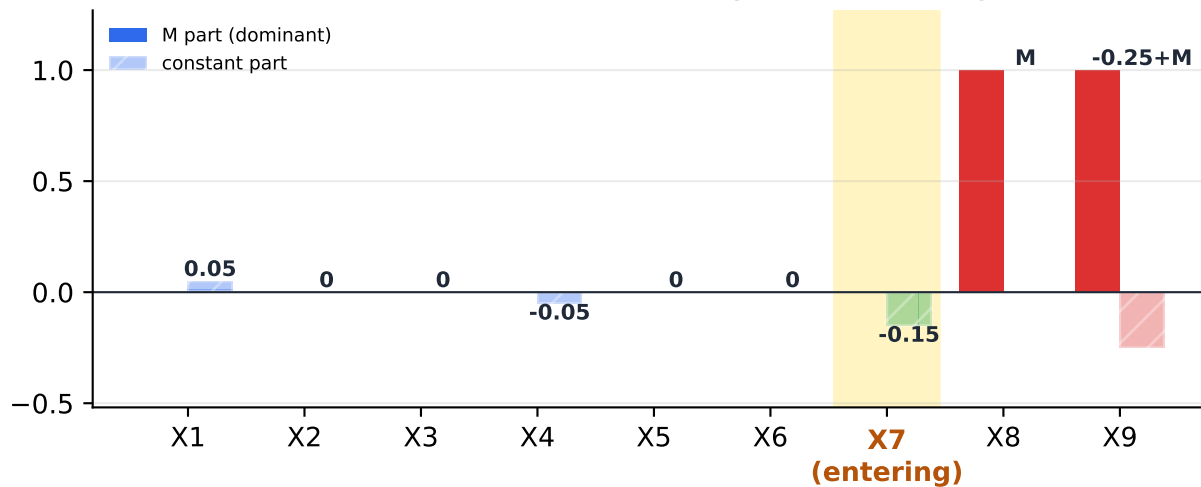
Row operations:

- Add $(-0.8333+3.3333M) \times R3$ to R0
- Add $3.3333 \times R3$ to R1
- Subtract $0.3333 \times R3$ from R2

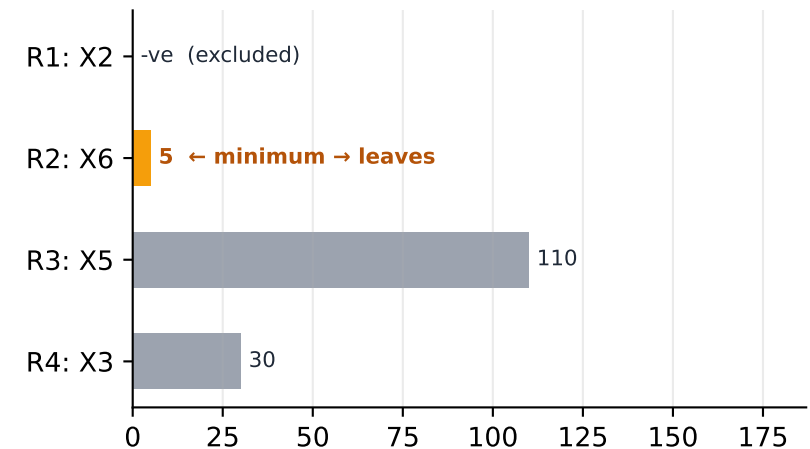
Iteration 4: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.05	0	0	-0.05	0	0	-0.15	M	-0.25+M	-29.5	
X2	1	0	1	1	0	1	0	0	-1	0	1	70	-ve
X6	2	0	-0.05	0	0	0.05	0	1	0.08	0	-0.1	0.4	5
X5	3	0	0.1	0	0	0.2	1	0	0.1	-1	0.3	11	110
X3	4	0	0	0	1	0	0	0	1	0	0	30	30

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X7-column)



Iteration 4 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X7 - it has the largest (in absolute value) negative coefficient in row 0 (-0.15).
- Minimum ratio test (RHS ÷ positive entries of the X7-column): R1: -ve, R2: 5, R3: 110, R4: 30.
- Leaving basic variable: X6 (row 2, smallest ratio). Pivot number = 0.08.

Iteration 4: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 0.08

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	0.05	0	0	-0.05	0	0	-0.15	M	-0.25+M	-29.5	
X2	1	0	1	1	0	1	0	0	-1	0	1	70	-ve
X6	2	0	-0.625	0	0	0.625	0	12.5	1	0	-1.25	5	5
X5	3	0	0.1	0	0	0.2	1	0	0.1	-1	0.3	11	110
X3	4	0	0	0	1	0	0	0	1	0	0	30	30

(b) X7 replaced X6 in the basis. Z = -28.75

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	-0.0437	0	0	0.0437	0	1.875	0	M	-0.4375+M	-28.75
X2	1	0	0.375	1	0	1.625	0	12.5	0	0	-0.25	75
X7	2	0	-0.625	0	0	0.625	0	12.5	1	0	-1.25	5
X5	3	0	0.1625	0	0	0.1375	1	-1.25	0	-1	0.425	10.5
X3	4	0	0.625	0	1	-0.625	0	-12.5	0	0	1.25	25

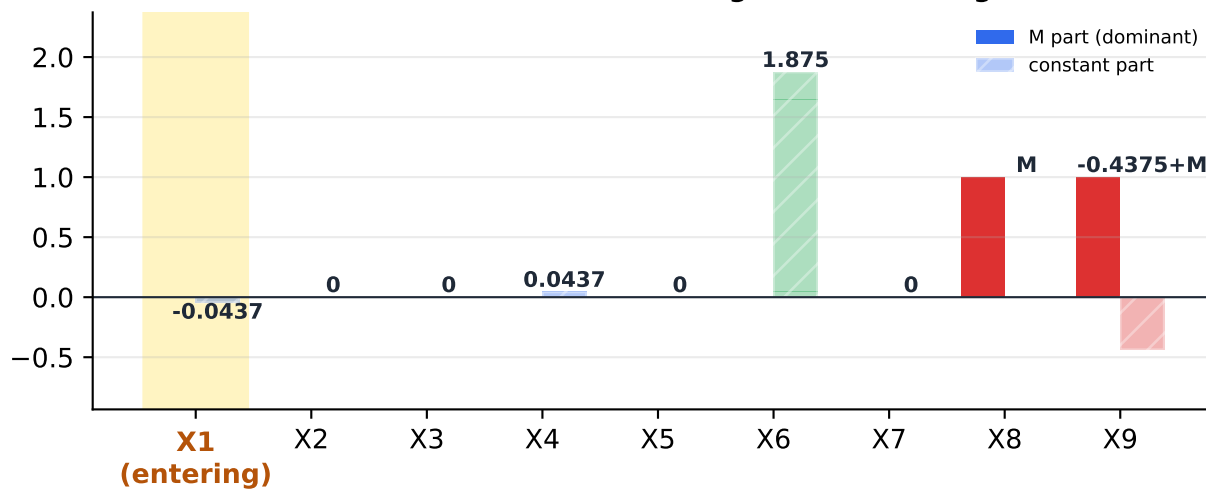
Row operations:

- Add $0.15 \times R2$ to R0
- Add $1 \times R2$ to R1
- Subtract $0.1 \times R2$ from R3
- Subtract $1 \times R2$ from R4

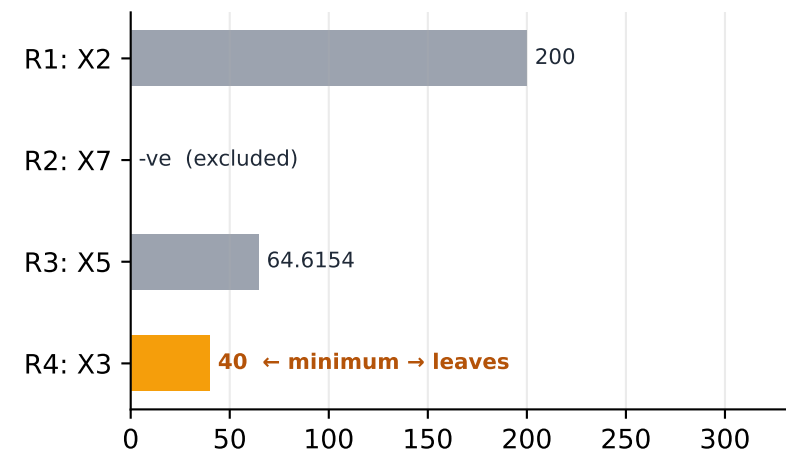
Iteration 5: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	-0.0437	0	0	0.0437	0	1.875	0	M	-0.4375+M	-28.75	
X2	1	0	0.375	1	0	1.625	0	12.5	0	0	-0.25	75	200
X7	2	0	-0.625	0	0	0.625	0	12.5	1	0	-1.25	5	-ve
X5	3	0	0.1625	0	0	0.1375	1	-1.25	0	-1	0.425	10.5	64.6154
X3	4	0	0.625	0	1	-0.625	0	-12.5	0	0	1.25	25	40

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X1-column)



Iteration 5 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X1 - it has the largest (in absolute value) negative coefficient in row 0 (-0.0437).
- Minimum ratio test (RHS ÷ positive entries of the X1-column): R1: 200, R2: -ve, R3: 64.6154, R4: 40.
- Leaving basic variable: X3 (row 4, smallest ratio). Pivot number = 0.625.

Iteration 5: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R4 by the pivot number 0.625

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS	Ratio
Z	0	1	-0.0437	0	0	0.0437	0	1.875	0	M	-0.4375+M	-28.75	
X2	1	0	0.375	1	0	1.625	0	12.5	0	0	-0.25	75	200
X7	2	0	-0.625	0	0	0.625	0	12.5	1	0	-1.25	5	-ve
X5	3	0	0.1625	0	0	0.1375	1	-1.25	0	-1	0.425	10.5	64.6154
X3	4	0	1	0	1.6	-1	0	-20	0	0	2	40	40

(b) X1 replaced X3 in the basis. Z = -27

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0	0	0.07	0	0	1	0	M	-0.35+M	-27
X2	1	0	0	1	-0.6	2	0	20	0	0	-1	60
X7	2	0	0	0	1	0	0	0	1	0	0	30
X5	3	0	0	0	-0.26	0.3	1	2	0	-1	0.1	4
X1	4	0	1	0	1.6	-1	0	-20	0	0	2	40

Row operations:

- Add $0.0437 \times R4$ to R0
- Subtract $0.375 \times R4$ from R1
- Add $0.625 \times R4$ to R2
- Subtract $0.1625 \times R4$ from R3

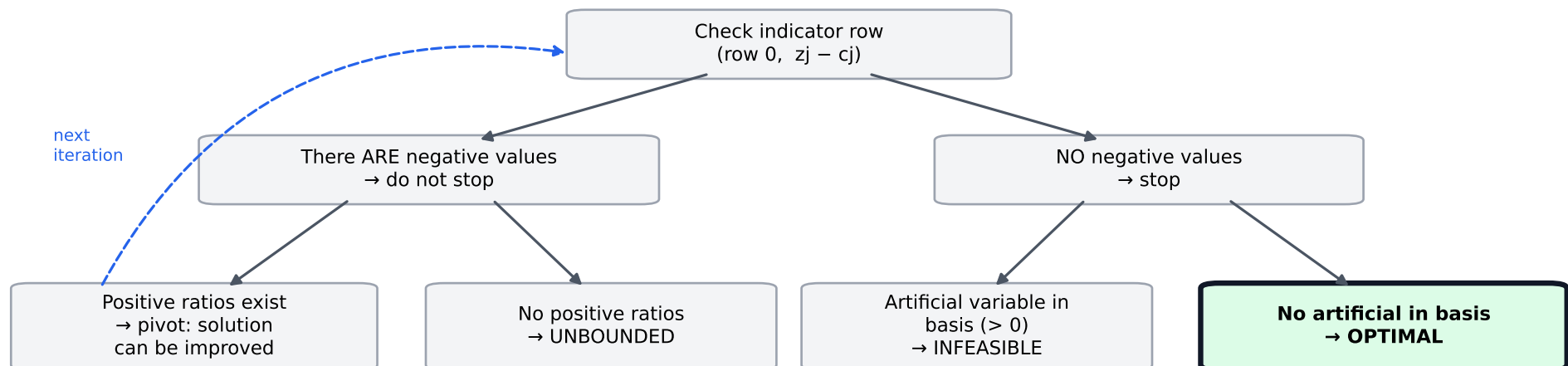
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	X9	RHS
Z	0	1	0	0	0.07	0	0	1	0	M	-0.35+M	-27
X2	1	0	0	1	-0.6	2	0	20	0	0	-1	60
X7	2	0	0	0	1	0	0	0	1	0	0	30
X5	3	0	0	0	-0.26	0.3	1	2	0	-1	0.1	4
X1	4	0	1	0	1.6	-1	0	-20	0	0	2	40

Final check.

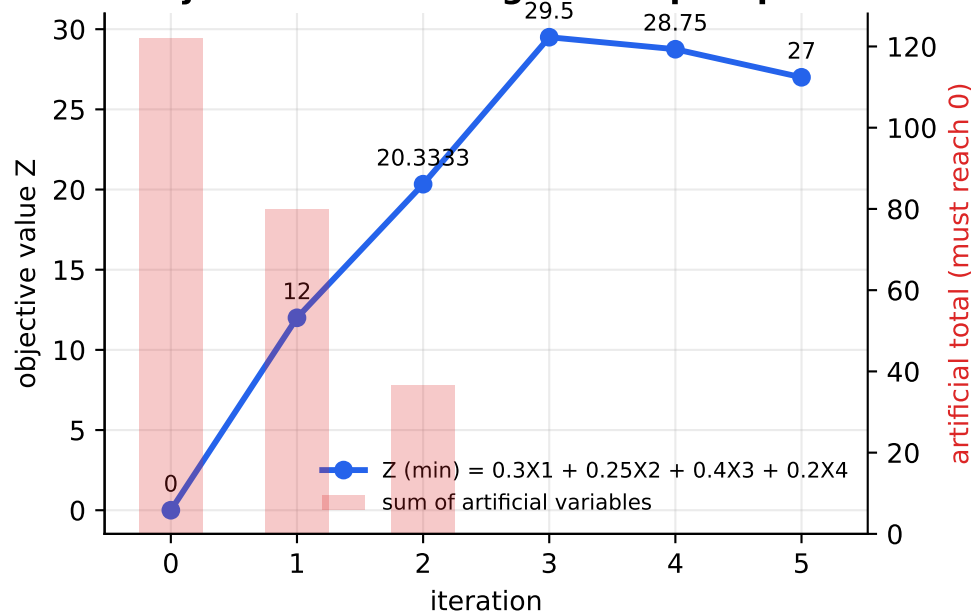
- Optimal: row 0 has no negative coefficients. $Z^* = 27$ (converted back: $Z = -Z'$)
- Alternative optima: non-basic X4 has a zero coefficient in row 0 - bringing it in changes the solution but not Z.

Simplex decision diagram - outcome: OPTIMAL

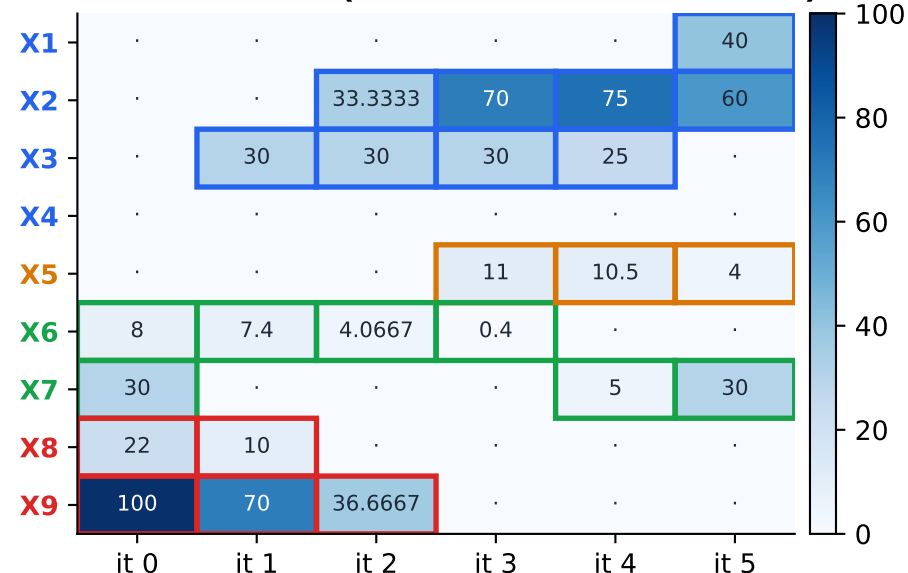


Practice - Four-Ingredient Blending (Minimize Cost) - OPTIMAL, $Z^* = 27$

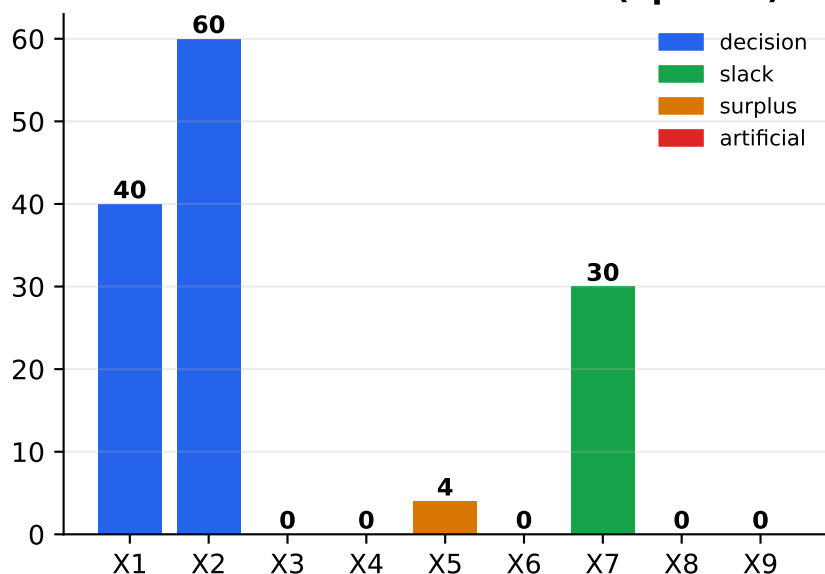
Objective value along the simplex path



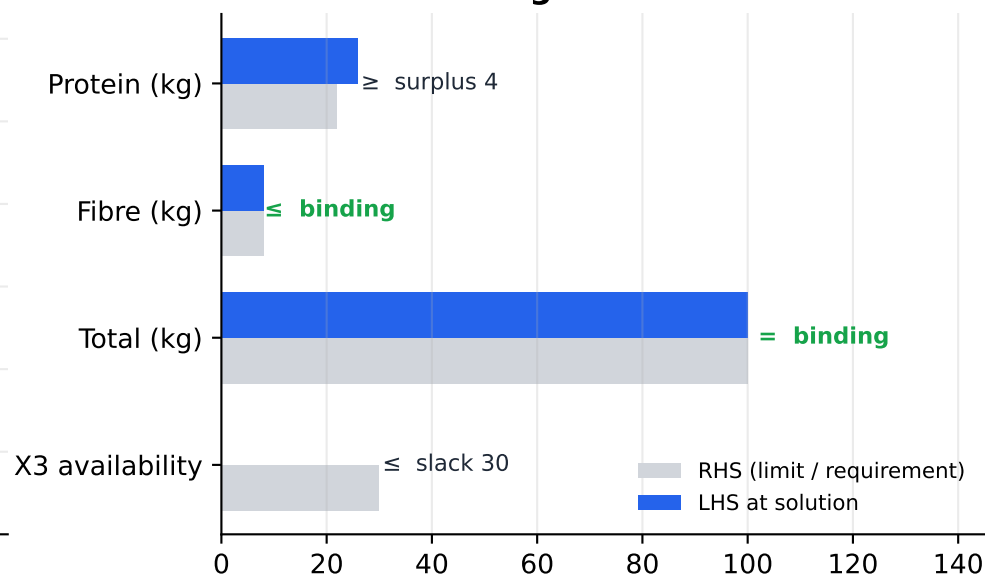
Basis evolution (outlined = basic variable)



Final values of all variables (optimal)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	27	27
X1	40	53.3333
X2	60	33.3333
X3	0	0
X4	0	13.3333

✓ Results agree